7.1 Assignment Operators

[This section corresponds to K&R Sec. 2.10]

The first and more general way is that any time you have the pattern

```
v = v op e
```

where *v* is any variable (or anything like `a[i]`), *op* is any of the binary arithmetic operators we've seen so far, and *e* is any expression, you can replace it with the simplified

```
v op= e
```

For example, you can replace the expressions

```
i = i + 1
j = j - 10
k = k * (n + 1)
a[i] = a[i] / b
```

with

```
i += 1
j -= 10
k *= n + 1
a[i] /= b
```

In an example in a previous chapter, we used the assignment

```
a[d1 + d2] = a[d1 + d2] + 1;
```

to count the rolls of a pair of dice. Using `+=`, we could simplify this expression to

```
a[d1 + d2] += 1;
```

As these examples show, you can use the ``op='' form with any of the arithmetic operators (and with several other operators that we haven't seen yet). The expression, *e*, does not have to be the constant 1; it can be any expression. You don't always need as many explicit parentheses when using the *op*= operators: the expression

```
k *= n + 1
```

is interpreted as

```
k = k * (n + 1)
```

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